

Cells: Structure, Transport & Division

Answer sheet · 33 marks total

Section A — Formula triangle

Top cell = **I** (image size) Bottom-left = **M** (magnification) Bottom-right = **R** (real size)

[1 mark each, 3 marks total]

Section B — Straight substitution

B1. $M = I / R = 6 / 0.3$ (1) $M = \times 20$ (1)

B2. $M = I / R = 15 / 0.05$ (1) $M = \times 300$ (1)

Section C — Rearrange and solve

C1. $I = M \times R$ (1) $I = 500 \times 0.002$ (1) $I = 1 \text{ mm}$ (1)

C2. $R = I / M$ (1) $R = 4 / 800$ (1) $R = 0.005 \text{ mm}$ (1)

C3. $20 \mu\text{m} = 0.02 \text{ mm}$ (1) $I = M \times R = 50 \times 0.02$ (1) $I = 1 \text{ mm}$ (1)

Section D — Cell structures

D1. Any two of: cell wall, chloroplasts, permanent vacuole (1 mark each, 2 marks total)

D2. Any two of: cell wall, single DNA loop / plasmids (note: bacterial cells do NOT have a nucleus, mitochondria or chloroplasts — do not accept these) (1 mark each, 2 marks total)

D3. Site of respiration, which releases energy for the cell (1)

D4. Site of photosynthesis (1)

Section E — Transport across membranes

E1. The net movement of particles (1) from a region of higher concentration to a region of lower concentration (1).

E2. The movement of water (1) from a region of higher water concentration to lower water concentration, through a partially permeable membrane (1).

E3. Mineral ions move from a lower to a higher concentration in the root hair, i.e. against the concentration gradient (1); this does not happen naturally, so energy from respiration is needed to move them (1).

Section F — Cell division & differentiation

F1. Growth (1) and repair of damaged tissue (1).

F2. Body cell = 46 (1); gamete = 23 (1); because meiosis halves the chromosome number, so that fertilisation (joining two gametes) restores the normal number, 46 (1).

Marking note: award method marks for a correctly rearranged equation and correct substitution even if the final answer contains an arithmetic error. In Sections D–F, accept any scientifically correct equivalent wording.